Environmental scanning, futures research, strategic foresight and organizational future orientation: a review, integration, and future research directions

René Rohrbeck*

Aarhus School of Business, Aarhus University, Fuglesangs Allé 4, 8210 Aarhus V, Denmark.
E-mail: rrohr@asb.dk

Manuel Bade

Christian-Albrechts-Universität zu Kiel, Westring 425, 24118 Kiel, Germany.
E-mail: mbade83@web.de

* Corresponding author

Abstract: In this paper we explore the current understanding on how firms explore future changes and trends as well as plan their managerial responses. We review literature in four research streams: (1) environmental scanning, (2) futures research, (3) peripheral vision, and (4) corporate/strategic foresight. Through the analysis of more than 250 articles we (a) trace the evolution over time, (b) highlight the linkages between the different research streams, and (c) give recommendations for future research. Overall we call for more cross-fertilization of the different research streams and a stronger linkage to adjacent research disciplines. Through such integration and linkage research should produce better recommendations for managers on how to build an organizational future orientation, drive organizational adaptation, and make their firms robust towards external discontinuous change.

Keywords: Environmental scanning, futures research, peripheral vision, strategic foresight, corporate foresight, dynamic capabilities, organizational ambidexterity, organizational adaptation.

Introduction

In the last four decades the predictability of future developments have decreased dramatically. Globalization led to the entrance of unknown competitors, the emergence of the Internet has created global markets and now serves as a fast and inexpensive marketing and distribution channel, risk capital allows small firms and new ventures to become global players in months, and social networks have the power to turn niche trends into huge markets within weeks. In consequence firms find it increasingly difficult to retain their competitive advantage and survive in these dynamic environments [1, 2].

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To succeed in dynamic environments firms can follow two major approaches: (1) through planning and (2) through structural means. *Structural means* include, (a) *diversification* in different business areas, thus balancing risk in one field with opportunity in the other, (b) *corporate venturing schemes*, where the firm is creating internal start-up companies and is counting on serendipity to bring out at least some very successful new businesses [3, 4], and (c) *ambidextrous organizational design*, where part of the company is focusing on managing the current business, while other parts explore new business that might profit from external change [5, 6].

In this paper we focus however on the first alternative, i.e. enhancing planning, that starts with identifying threats and opportunities early and subsequently translates these future insights into managerial action. Our starting point is the proposition put forward by Igor Ansoff that disruptive change can be anticipated by searching for weak signals, interpreting these and triggering organizational responses [7]. While Ansoff's work was mostly concerned about strategic planning and management, we also expect that other functions of the firm need to engage into future-related planning. Particularly marketing has a role in spotting changes in customer behavior and needs [8] and innovation management can for example plan future product generations on the basis of market opportunities [8, 9].

Research on this future related planning has been conducted within different research streams. Early work was primarily using the term *environmental scanning* emphasizing the need for active search and the environment as the search target [10, 11]. Under the term *futures research* scholars build primarily on the logic that the future is and will remain uncertain, thus future-oriented planning should aim to explore possible futures rather than trying to predict the one future [12, 13]. In the 2000s there was also some research under the term *peripheral vision*, emphasizing similarly to environmental scanning, that signals on discontinuous change need to be spotted outside the current business and that firms need to build specific sensors to detect it [14, 15]. In recent years the term *strategic/corporate foresight* has gained in popularity and will be regarded as a fourth research stream in our analysis. In this latter research stream the focus is on the process of translating future insights into managerial actions [2, 16].

Methodology

Our overall research aim is to explore the current understanding on future-related planning activities that are used to build a future orientation in the firm. In that respect we define:

> *Organizational future orientation (OFO) is the ability to identify and interpret changes in the environment and trigger adequate responses to ensure long-term survival and success.*

Our research aim can be broken down into three sub-issues:

- Identify key topics and findings within each research stream
- Describe the relationship between the research stream
- Give recommendations for future research

To identify relevant literature we use the Web of Science® (WoS) database that contains all journals in the Social Science Citation Index (SSCI). That gives us a sufficiently broad range of articles and ensures also a sufficient quality as SSCI listed journals are all double-blind peer-reviewed and are among the leading journals in their research fields.

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As keywords for our search we used the four research streams: (1) environmental scanning, (2) futures research, (3) peripheral vision and (4) corporate/strategic foresight. The latter one has been entered into the search as two keywords 'corporate foresight' and 'strategic foresight'. We have decided to combine these two because an initial review of articles has shown that the perspectives are similar. The only difference is that corporate foresight is focusing exclusively on the firm perspective, while under strategic foresight a high proportion of articles is discussing the perspective of national foresight activities, for example to identify which future technologies yield the highest societal welfare potential and thus which research should be supported by governmental subsidies and tax incentives.

Time-wise we decided to focus on the time after the early seminal work from Igor Ansoff, as we could only identify very few other articles before 1980 and our research is also aimed primarily at identifying the state-of-the-art and not trace back the historic roots of the respected research streams. The last articles that we included were the once published in November 2011.

To identify the relevant articles we used two filter, as can be seen in Figure 1. In the first filter we used the keywords plus the subject area "business economics" in the WoS database. This subject area includes all topics that are related to management science and helped us thus to exclude all articles that had were discussing for example results form foresight exercises in particular fields, thus not dealing with advancing the research on foresight, but research in the particular field (for example the future of biotech). Through this automated search and filtering we identified 259 articles.

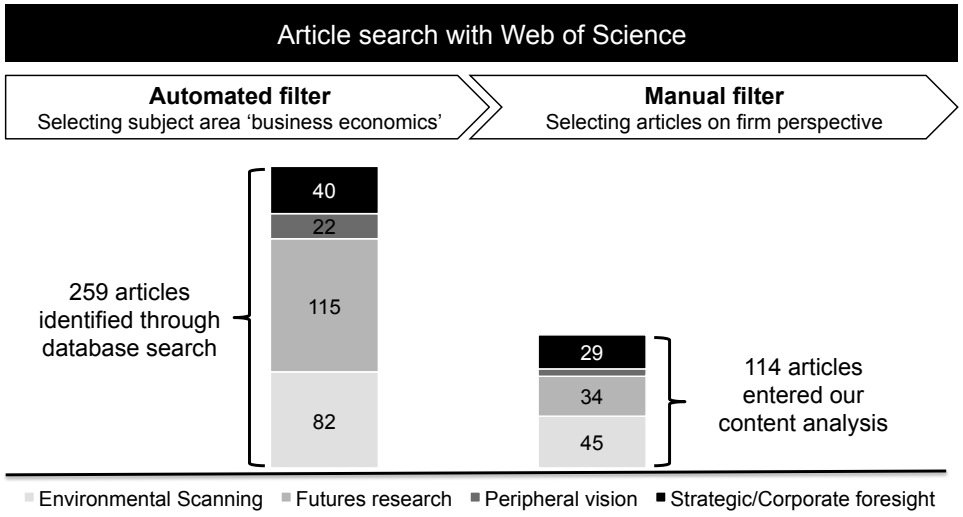


Figure 1: Search process on the basis of the Web of Science database and manual filtering

This set of 259 however still included articles that related to future-related planning from a national perspective, while we aim our analysis exclusively on the firm perspective. To identify the articles on the firm perspective we reviewed the papers manually. In most cases the abstract was already sufficient; in only a few reading through the whole paper was required. At the end of this second filtering we had a set of 114 articles in total. The larges proportion was with 45 papers the environmental scanning

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field, followed by futures research with 34, corporate/strategic foresight with 29, and peripheral vision with 6 papers.

These 114 articles entered our content analysis, where all articles were reviewed, classified concerning their methodology, empirical basis (where applicable), and major findings. In addition the detailed review of the article content formed the basis of the discussion of the evolution of the different research streams over time and the discussion of the relationships between the research streams.

Literature on how firms explore their futures

When plotting the articles on a time scale, see Figure 2, we can see that the early research was conducted only under the two terms environmental scanning and futures research. The *environmental scanning* articles followed mostly the conceptual propositions form Igor Ansoff [7] and tried to establish if and to what ends environmental scanning was used [17, 18]. The articles were also predominantly empirical and often based on quantitative analysis of survey data. The *futures research* articles where mostly conceptual or literature reviews that aimed to build the frame on how to integrate the 'multiple-futures'-logic into corporate and strategic planning [12, 19]. Another issue was how to motivate the adoption of futures research and how to motivate for participation in futures-research exercises [20, 21].

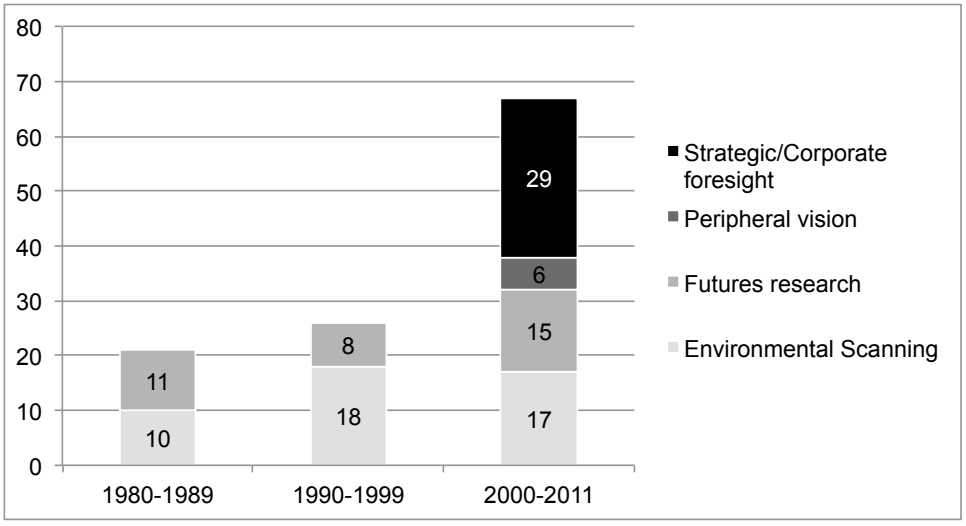


Figure 2: Development over time, number of articles published in the respective research streams in the years 1980 – 2011.

The research streams peripheral vision and corporate/strategic foresight emerged as important research streams only after 2000. The most influential authors in the peripheral vision research stream are Day and Schoemaker, who have also created the term for a conference and a special issue in the Long Range Planning journal. The emergence of the research stream corporate/strategic foresight has been driven by a larger community of scholars and has been growing much faster than the others.

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In the following we want to discuss more into detail the major topics covered in the different research streams.

Environmental Scanning

In an early work Hambrick proposes that environmental scanning activities should be employed to focus more attention on trends and changes in the firm´s environment in order to learn from these events [22]. Jain emphasizes that this is particular important for large firms that need to deal with a rising complexity, both in the environment and through the increase of complexity of internal structures. This raise complexity is increasing the need for a systematic environmental scanning [23]. Jain further emphasizes that firms need to scan not only in the technological but also in the social, economic, and political environment [23]. Sawyer adds the uncertainty dimensions to the reasons why firms need environmental scanning. He finds that scanning activities of firms rise with the level of uncertainty perceived by its top management [24].

To enhance environmental perception, top management functional diversity is found to have a positive impact [25]. Also specific techniques can be used such as the quick environmental scanning technique of Nanus [26, 27] or the on-line computerized data base search [28, 29]. Overall the aim is to use environmental scanning to enhance the firm's performance [30].

It has also been shown that even for small firms environmental scanning seems to pay off [31]. For small firms, that can allocate less resources on formal environmental scanning activities or invest into dedicated units, it is suggested that maintaining a personal information network can be a good alternative [32]

Futures Research

The starting point for futures research is the paradigm that corporate planning under uncertainty needs to move away from forecasting and predicting towards identifying multiple possible futures [19, 33, 34]. These can than be used to plan flexible strategies, increase strategic agility or assess the robustness of the firm's strategy.

In the 1980 research contributions where primarily focused on establishing the "raison d'être", i.e. the potential usefulness of futures research. Eppink [12] linked it to corporate and strategic planning and emphasized the importance to continuously monitor changes in the environment. Morris [35] pointed out its usefulness for product-planning. Sims and Eden [20] emphasis the role of facilitating discussion in the management teams, including the creation of shared believes and dealing with conflicts. Becker sees futures research primarily as a tool to orchestrate and coordinate ideas from disperse sources. And finally Masini [36] points at its transformative power, if applied in a project like fashion, where desired futures are identified and later build together.

But there were also calls for increased focus of futures studies. Or as Amara [37] puts it:

In order to survive it [futures research] needs to dispense with its tendency to be 'all things to all people'

Linstone [38] also pointed already early at the difficulty to bridge the gap between analysis and action. Decision-makers and planners are just not ready to deal with multiple and often conflicting views of the future.

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In the 1990s the focus was mainly on advancing methods and processes of futures research [39, 40]. Helmer [41] suggests to use that more the new possibilities create through the Internet, by for example enabling the use of a worldwide Delphi-Net (in reference to the Delphi technique) that would help to support governmental and industrial planning. Coates [42] emphasizes that the awareness of the corporation´s external environment and long-term future is getting increasingly important in technology planning. He adds that it is important not to focus entirely on the firm, but to analyze the whole industry to identify factors that will drive its futures. A particularly interesting path, which is still pursued today, is also the usage of complexity theory in futures research [43, 44].

The call to connect futures research more with other management functions [45], has however remained largely unanswered, at least from futures research scholars.

Peripheral Vision

In contrast to futures research, the research stream on peripheral vision has from the outset been primarily concerned about the firm perspective. The basic proposition is that that firms need to look for information that lies beyond their core business. Scanning for technological and market developments is important for survival and retaining the long-term competitiveness.

Day and Schoemaker have put the term peripheral vision forward for their special issue in Long Range Planning of (2004, Volume 37, Issue 2). In addition to the special issue they also organized a conferece at the Wharton School under the same term. The conference brought together 100 practitioners and scholars. The special issue compiled a set of interesting articles from leading scholars, primarily from the field of strategic management.

Day and Schomaker assume that firms are organized and managed in a way that draws the attention of management towards the current business and the insight of the firm. This results into a systematic ignorance to changes and events outside the firm. They compare this situation to driving though fog and argue that special sensors are needed to counterbalance this systematic ignorance [46]. This view is further emphasized by the article from Sidney Winter in which she explains that firms tend to reemphasize the sensors that made them successful, leading to in inadequacy of sensors in a new and altered environment [47].

Heackel sees the need for new sensors to be routed even deeper. He argues that the nature of business has changed from a “make-and-sell” to a “sense-and-respond”-paradigm [8].

Prahalad points out, that organizations have an inherent dominant logic that keeps them on the road, but also may act as a blinder towards change. While the dominant logic is good for running the current business effectively and efficiently, it hinders the ability of adaptation to environmental changes. To overcome this blinder, firms should focus on next practices, use low cost experimentation, look beyond the borders of industries and beyond geographical borders [48]. Another way to prevent such peripheral blindness is to use teams for interpretation and ensure a sufficient diversity in team composition [49].

While the research under the term peripheral vision has generated important contributions to the overall research field, the term as such has lost appeal in the research community. The findings and proposition however continue to influence research today

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and are mostly followed up and referenced in the research stream on corporate/strategic foresight.

Corporate/Strategic Foresight

It can be argued that corporate/strategic foresight (CF) can at least be traced back to the propositions of Igor Ansoff, that strategic management will move towards real-time and competitive advantage will be build on the basis of timely responses to external (discontinuous) change [7]. Thus the argument for the need for CF capabilities is also based on the need for firms to adapt to external change to retain their competitive advantage [50]. In addition CF research has also draws on the methodology literature from the broader (national and societal) foresight field.

As we have argued earlier CF has also become a term of choice for scholars that have previously worked in the field of peripheral vision and futures research, giving it already the function of consolidating the field. This shift of terminology can also be seen in the increase of articles that are being published under the term corporate foresight. In Figure 2 CF research accounts already for over 40% of the overall field, while in the early phases from 1980-1990 and from 1990-2000 none of the articles was published under the CF term.

The research on CF builds on the proposition that firms need a continuous process for identification and interpretation of change and triggering adequate organizational responses [2, 51, 52]. In such a process the identification of weak signals and their filtering and interpretation is crucial [53, 54]. It has also been noted that particularly the interpretation should be done through top management as they are the only ones in a firm that have a sufficiently high vantage point [55].

The outcome of CF activities can be an enhanced market intelligence [56-58], enhances innovation capacity [9, 59, 60], enhanced competitive intelligence [61], the identification of promising new business fields [16] and better strategic decision-making [62-64].

A particular emphasis is put on the identification and management of wild cards, i.e. events that are singular, sudden, surprising and shattering (serious and severe consequences). A CF system or approach should thus be able to incorporate these rare but high impact events into planning and in case of occurrence trigger the appropriate managerial responses in a timely fashion [65-68].

The research stream has also produced a good collection of case studies of CF practices, which reveal methods [69-72], approaches [73] as well as the organizational outcomes [56, 57, 74]. Overall CF is expected to facilitate organizational interpretation [75] and learning [76] and thus can be one of the key element to ensure organizational adaptation in times of environmental change.

An integrated perspective on organizational future orientation

In Table 1 we have attempted to compare the different research streams concerning their basic proposition, i.e. their dominant logic, the tools and techniques and the desired outcome they seek.

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	Environmental scanning	*Futures research*	*Peripheral vision*	*Corporate/ strategic foresight*
Basic proposition	(Top) management needs to scan the environment in order to respond to external change	The best way to deal with uncertainty is to plan on the basis of multiple possible futures	Firms need to develop specific sensors to detect change in the periphery early in order to be able to retain their competitive advantage	Firms need to build integrated systems that allow translating perceived change into managerial action
Tools and techniques	Environmental scanning by (top) management, by networking or active search for weak signals	Methods that build on systems logic and allow to deal with multiple futures	Focus on the establishment of sensors that can be build on methods, processes or people	A central process that utilizes foresight methods and that is linked to other functional units such as strategic and innovation management
Desired outcome	Enhanced firm performance	Plan under uncertainty and influence/create the future	Fast response to environmental change	Enhanced responsiveness towards external change, particular for enhancing strategic decision-making and the innovation capacity

Table 1: Comparison of the different research streams

Conclusion

By reviewing more than 30 years of scholarly research on organizational future orientation we hope to contribute to the integration of the various research streams. In addition we use the different lenses, provided by the research streams, to highlight their major contributions to the overall understanding on how firms should deal with their futures. By comparing them on their basic assumptions, their tools & techniques and their desired outcomes, we wish to provide a basis for further integration of the research field.

As we have shown in Figure 2, the development over time seem already to point at an integration of the different research streams under the common term corporate/strategic foresight. In this integration it will be important to build on the knowledge on how firms scan for change in their environment (environmental scanning perspective), how they plan by using multiple possible futures, rather than predictions (futures research perspective) how firms can build adequate sensors (peripheral vision perspective) and how to tie these elements into and integrated organizational response system.

From the conceptual work of Daft and Weick [75] we know that an organizational response might be generated along three phase, scanning (data collection), interpretation

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(data given meaning), and learning (action). However the empirical evidence, particularly from the corporate/strategic foresight research seems to suggest that firms still lack systems that ensure that these steps can be taken repeatedly in order to ensure that all relevant external change is perceived and response are triggered [74, 76, 77].

Future research directions

Integration with other research disciplines

In the discussion of the research streams we have already commented on their relationship to broader research disciplines such as strategic and innovation management. While environmental scanning has its roots and has always kept a direct link to strategic management, corporate/strategic foresight is often referencing towards innovation management literature. This latter link has its roots in the close relationship between technology foresight and technology/R&D/engineering/innovation management. Another link from foresight can also be traced towards marketing and particularly exploratory marketing. Some reference foresight also to risk management, particularly concerning discontinuous change and wild cards that translate into major threats for the firm's survival.

Futures research and peripheral vision have much less connection to other major disciplines and are thus more reliant on their own basic theoretical propositions. This is a lack which has been pointed out before [45, 78], but too few scholars have responded to these calls.

We believe that particularly the strategic management theoretical frameworks can be used to advance the field of organizational future orientation. Four theoretical frames can be expected to be particularly useful: dynamic capabilities [79], strategic agility [80], organizational ambidexterity [81, 82] and strategic adaptation [50]. Within the innovation management discipline the areas radical innovation [83] and (technological) disruptions [84] can be expected to have a close link.

Making these links would provide the organizational future orientation perspective with powerful meta-frameworks. And in the other direction organizational future orientation can advance the theoretical frameworks by contributing understanding about the underlying micro processes.

Jointly building an understanding of Organizational Future Orientation

In contrast to economics the management research field is lacking a meta-model that ties the different research streams together. This leads to a large variety of research topics that often explore similar phenomena, but use different terms to describe them. If these research streams stay ignorant towards the related or similar activities we run the risk to continuously "re-invent the wheel", fail to build on each other's work and end up giving conflicting recommendations to managers.

As we have shown in the analysis of the different research streams this is already happening in the field that we choose to call organizational future orientation. The problem is amplified by the lack of reference to meta-frameworks such as dynamic capabilities (to just name one).

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We believe that more effort is required to consolidate our terminology and theoretical frameworks as well as to increase the cross-referencing between the different research streams. It would also do a lot of good for advancing our research field, if we were to conduct more studies to test and reproduce earlier findings rather than aiming to identify and explore yet another phenomenon that lacks a connection to earlier findings.

Many research disciplines have profited from developing a meta-frame of reference. One example is the new product development (NPD) best-practice framework, which defines the maturity of a firm's innovation management system [85]. The Organizational Future Orientation field has also already two attempts to create such a maturity model: Terry Grim's "Foresight Maturity Model" [86] and René Rohrbeck's "Maturity Model of Organizational Future Orientation [87].

I believe that both models are not a final answer to the call for a suitable meta-frame for our field, but they should provide a sound basis for discussion and are certainly an open invitation to other scholars to join in the development.

We hope that many will join in this quest to mature our research field and link it to the established management research disciplines.

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